

4. Footing segment

| No | Design Parameter                  | Symbol             | Value           | Source           |
|----|-----------------------------------|--------------------|-----------------|------------------|
| 1  | Total factored axial load         | P <sub>u</sub>     | 1609.756 kip    | Main Tank Design |
| 2  | Factored overturning moment       | M <sub>u</sub>     | 5673.437 kip-ft | Main Tank Design |
| 3  | Factored shear                    | V <sub>u</sub>     | 52.30 kip       | Main Tank Design |
| 4  | Number of legs                    | N <sub>leg</sub>   | 4               | Tank Geometry    |
| 5  | Pedestal size                     | B × L              | 39 × 39 in      | Pedestal Design  |
| 6  | Pedestal height                   | H <sub>p</sub>     | 7'-5"           | Pedestal Design  |
| 1. | Concrete strength                 | f <sub>c</sub>     | 4 ksi           | User Input       |
| 8. | Reinforcing steel                 | f <sub>y</sub>     | 60 ksi          | User Input       |
| 9. | Allowable soil pressure (Assumed) | q <sub>allow</sub> | 3.0 ksf         | User Input       |

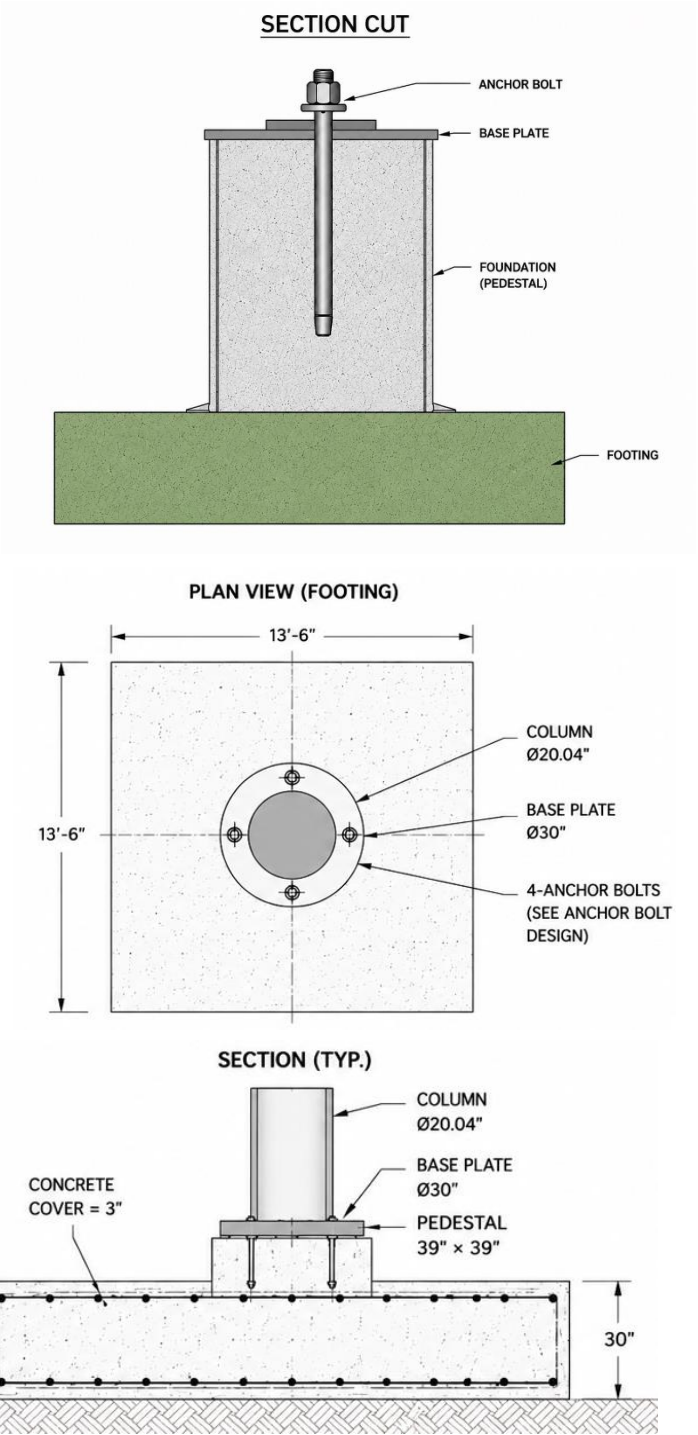
Step 1 – Footing Load

Gravity load per footing

Compression Side

$$P_{Leg} = \frac{P_u}{4} = 402.44\text{kip}$$

$$P_c = P_{leg} + 103.65$$



$$P_c = 506.09 \text{ kip}$$

Use

$$P = 520 \text{ kip}$$

## Step 2 – Preliminary Footing Size

Required footing area

$$A = \frac{P}{q_{allow}}$$
$$A = \frac{520}{3} = 173.3 \text{ ft}^2$$

Adopt

$$13.5 \times 13.5 \text{ ft}$$

Area

$$182.25 \text{ ft}^2$$

Which exceeds the required area.

## Step 3 – Soil Bearing Check

Applied pressure

$$q = \frac{520}{13.5 \times 13.5}$$
$$q = 2.85 \text{ ksf}$$

Allowable

$$q_{allow} = 3.0 \text{ ksf}$$

$$2.85 \text{ ksf} < 3.0 \text{ ksf} \dots \text{Pass}$$

## Step 4 – Footing Thickness

Use: 30 inch

This thickness is adequate for preliminary punching shear and one-way shear and is commonly used for elevated steel water tank foundations of this capacity.

## Step 5 – Flexural Reinforcement

Provide

Bottom reinforcement

$$\#8@8'' \text{ each direction}$$

Top reinforcement

#6@12" each direction

Concrete cover = 3inch

### Step 6 – Sliding Check

Use

$$FS = \frac{\mu P}{V}$$

Take

$\mu = 0.50$

$$FS = \frac{0.50(520)}{13.075}$$

$$FS = 19.9 > 1.5 \dots \dots \dots Pass$$

### Step 7 – Two-Way Punching Shear Check

Given:

- Footing size =  $13.5 \times 13.5$ ft
- Footing thickness,  $h = 30$ in
- Pedestal =  $39 \times 39$ in
- $f'_c = 4.0$ ksi
- $f_y = 60$ ksi
- Factored pedestal load used for design:

$$P_u = 520 \text{ kip}$$

- Factored soil pressure:

$$q_u = \frac{520}{13.5(13.5)} = 2.85 \text{ ksf}$$

- Bottom reinforcement = #8 @ 8 in each way
- Concrete cover = 3 in

### 7.1 Effective Depth

For #8 bars:

$$d = h - \text{cover} - \frac{d_b}{2}$$

Using #8 bar diameter  $d_b = 1.0\text{in}$ :

$$d = 30 - 3 - \frac{1.0}{2}$$

$$\mathbf{d = 26.5\text{ in}}$$

## 7.2 Critical Punching Shear Perimeter

For an approximately square pedestal, the critical section is taken at approximately  $d/2$  from the pedestal face.

Distance from pedestal face to critical section:

$$\frac{d}{2} = \frac{26.5}{2} = 13.25\text{ in}$$

Therefore, the critical-section dimensions are:

$$c_1 + d = 39 + 26.5 = 65.5\text{ in}$$

$$c_2 + d = 39 + 26.5 = 65.5\text{ in}$$

Thus,

$$\mathbf{b_o = 4(65.5) = 262\text{ in}}$$

or

$$\mathbf{b_o = 21.83\text{ ft}}$$

## 7.3 Area Inside Critical Perimeter

$$A_o = \frac{65.5(65.5)}{144}$$

$$\mathbf{A_o = 29.79\text{ ft}^2}$$

The soil inside the punching perimeter provides upward resistance.

$$P_{soil} = q_u A_o$$

$$P_{soil} = 2.85(29.79)$$

$$\mathbf{P_{soil} = 84.9\text{ kip}}$$

## 7.4 Factored Punching Shear

The punching shear force is:

$$V_u = P_u - P_{soil}$$

$$V_u = 520 - 84.9$$

$$V_u = 435.1 \text{ kip}$$

## 7.5 Punching Shear Stress

$$v_u = \frac{V_u}{b_o d}$$

Using inches:

$$v_u = \frac{435.1(1000)}{262(26.5)}$$

$$v_u = 62.7 \text{ psi}$$

or

$$v_u = 0.063 \text{ ksi}$$

## 7.6 Concrete Punching Shear Capacity

For an interior square pedestal with no shear reinforcement, a conservative ACI-based expression is:

$$v_c = 4\sqrt{f'_c}$$

With:

$$f'_c = 4000 \text{ psi}$$

$$v_c = 4\sqrt{4000}$$

$$v_c = 252.98 \text{ psi}$$

Using:

$$\phi = 0.75$$

the design capacity is:

$$\phi v_c = 0.75(252.98)$$

$$\phi v_c = 189.7 \text{ psi}$$

## 7.7 Punching Shear Check

$$v_u \leq \phi v_c$$

$$62.7 \text{ psi} < 189.7 \text{ psi}$$

**PASS**

The punching shear utilization is:

$$\frac{v_u}{\phi v_c} = \frac{62.7}{189.7} = 0.33$$

**Utilization = 33%**

Therefore, the 30-in-thick footing has substantial punching shear capacity.

### Step 8 – Punching Shear Capacity in Force Form

We can also show the check directly in terms of force.

$$V_c = v_c b_o d$$

$$V_c = 252.98(262)(26.5)/1000$$

$$V_c = 1,757 \text{ kip}$$

Design capacity:

$$\phi V_c = 0.75(1757)$$

$$\phi V_c = 1,318 \text{ kip}$$

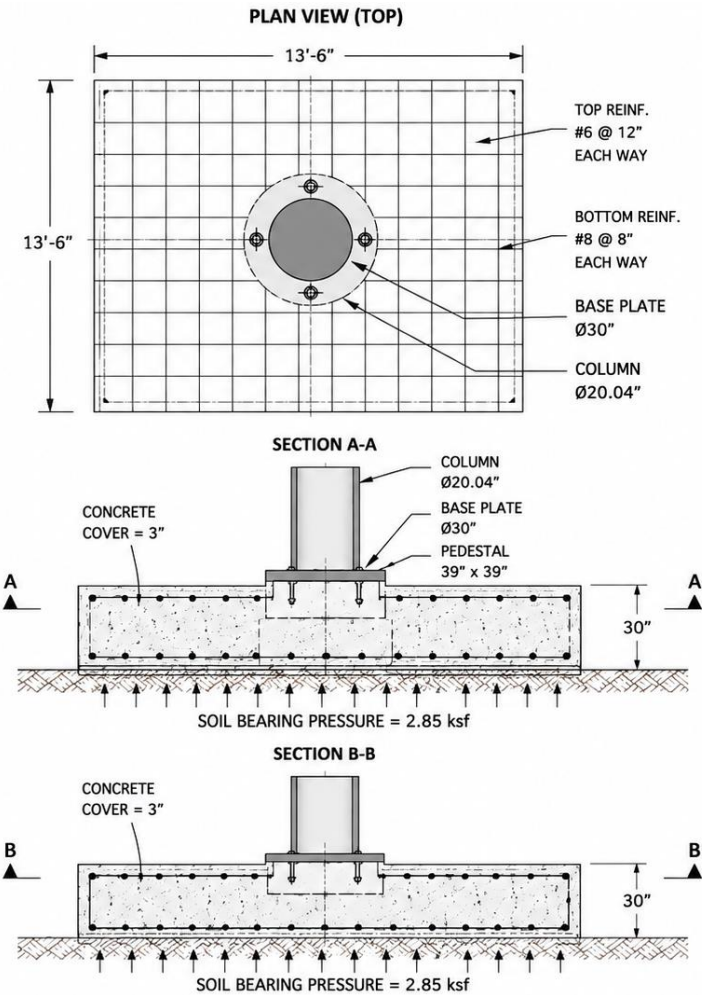
Compare:

$$V_u = 435.1 \text{ kip}$$

$$\phi V_c = 1318 \text{ kip}$$

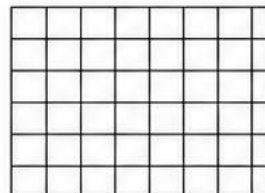
Therefore:

$$435.1 < 1318 \text{ kip} \text{ — PASS}$$

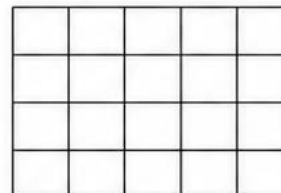


### REINFORCEMENT DETAILS

BOTTOM REINFORCEMENT  
#8 @ 8" EACH WAY



TOP REINFORCEMENT  
#6 @ 12" EACH WAY



### COLUMN & BASE PLATE DETAIL

